
The Myth of NHL Trap Games: A Systematic Analysis

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Abstract

Trap games in hockey are situations where a strong team underperforms against a weaker opponent after a more challenging win. Despite the popularity of this narrative, we find limited systematic evidence over 18 NHL seasons. We formalize a definition of trap games, analyze performance differences in trap games, build permutation tests, and evaluate multivariate models. We find limited evidence that certain teams underperform in trap games, however these results are not robust under multiple hypothesis correction. Trap games have minimal impact when predicting game outcomes. While memorable trap games that fit this narrative occur, there is no systematic pattern.

1. Introduction

A trap game in hockey is a situation where a heavily favoured team unexpectedly loses to a much weaker opponent. The favoured team has typically just won against a strong opponent in a preceding game that was emotionally-draining and high-intensity, which in theory may lead them to underestimate their weaker opponent.

In this work, we will build a systematic way to identify trap games, analyze performance trends, build a permutation testing framework, and examine feature importance in multivariate models.

2. Identifying Trap Games

A publicly available database (NaturalStatTrick, 2025) contains detailed records of each NHL game played between the 2007-08 season and the 2024-25 season. We pulled statistics for each game across all situations rather than solely on 5v5 or other special situations as a trap game is inherently linked to human psychology and humans do not care whether they outscored an opponent on 5v5 if they still lost the game.

¹All work, code, and views expressed here are personal and unaffiliated with my employer.

For a game i , let $g_{i,t}$ denote the number of goals scored by the team t and $g_{i,o}$ be the number of goals scored by their opponent. The number of points earned by the team during the game can thus be defined as:

$$p_{i,t} = \begin{cases} 2 & \text{if } g_{i,t} > g_{i,o} \\ 1 & \text{if } g_{i,t} = g_{i,o} \\ 0 & \text{if } g_{i,t} < g_{i,o} \end{cases}$$

Let $n_{i,t}$ be the number of games the team t has played during the season, up to and including the game i . Thus, we derived the rolling average number of points per game in the season to date, excluding game i , as:

$$a_{i,t} = \frac{1}{n_{i,t} - 1} \sum_{k=1}^{n_{i,t}-1} p_{k,t}$$

An NHL season is 82 games long and during the first part of the season, the average number of points earned in a game was highly volatile. Teams were more likely to get off to a hot or cold start and thus their average performance across a small sample size was not representative of a team's true talent level. Therefore, we only started categorizing teams around the quarter-season mark after game-by-game volatility decreased and teams were better aligned with their true talent level.

Specifically, we defined a team t as strong if $a_{i,t} > 1.05$ where $i > 20$. On the other hand, a team t was defined as weak if $a_{i,t} < 0.95$ where $i > 20$. These thresholds were empirically derived to represent approximately 30% of all teams each, thus balancing teams between strong, weak, and in-between.

For each game i we were then able to tag if the game was a trap game for team t using the following methodology:

$$R_{i,t} = \begin{cases} 1 & \text{if } (a_{i,t} > 1.05) \wedge (a_{i,o} < 0.95) \\ & \wedge (p_{i-1,t} = 2) \wedge (a_{i-1,o} > 1.05) \\ 0 & \text{otherwise} \end{cases}$$

A trap game therefore has two criteria, that a strong team was facing a weak team and the strong team won its previous

Table 1. Overall statistics for Trap Games vs Similar Non-Trap Games.

CATEGORY	# GAMES	GF	GA	GF%
TRAP GAME	1,000	3.3	2.6	57.0
SIMILAR NON-TRAP GAMES	4,330	3.2	2.6	55.8
% CHANGE		2.7%	-0.2%	2.1%

Table 2. % Change in trap games vs similar non-trap games over time.

SEASON BUCKET	GF	GA	GF%
PRE-LOCKOUT (2007–2012)	-0.0%	-2.7%	3.3%
POST-LOCKOUT (2013–2019)	0.5%	2.5%	0.1%
PANDEMIC (2020–2021)	-3.2%	-11.4%	4.3%
POST-PANDEMIC (2022–2025)	6.1%	0.7%	2.1%

game against another strong team. Approximately 2.3% of all games were tagged as trap games using this approach.

3. Performance Trends in Trap Games

First, we defined a similar non-trap game as a game played by a strong team against a weak opponent that does not otherwise fit the criteria of $R_{i,t}$.

Table 1 shows the strong team performs better in trap games when looking at the totality of our data across all seasons. In trap games, the stronger team on average scored 2.7% more goals and allowed a roughly equal number of goals as they did in similar non-trap games.

This trend holds across the majority of NHL seasons, as seen in Table 2. However, in the post-lockout period between 2013 and 2019, we observed that opposing teams scored more in trap games than they did in similar non-trap games, thus making them more equal.

While teams generally outperformed in trap games than they did in similar non-trap games, this was not the case for all individual teams. Table 3 shows ten teams who were often in trap game situations during the post-pandemic (2022–2025) time period.

Teams such as the Florida Panthers stood out as doing better in trap games than they did in non-trap games. On the other hand, teams such as the Carolina Hurricanes underperformed in trap games. Sample sizes for all teams in table 3 were in the range of 50 games.

Detailed results for all teams and seasons are on GitHub.

Table 3. % Change in trap games vs similar non-trap games over time for selected teams during the post-pandemic (2022–2025) era.

TEAM	GF	GA	GF%
FLORIDA	26.2%	-28.5%	21.3%
TAMPA BAY	6.5%	-26.9%	11.6%
TORONTO	17.7%	-0.7%	10.2%
BOSTON	17.6%	-2.5%	6.4%
WINNIPEG	13.8%	14.4%	-0.7%
DALLAS	-10.3%	0.1%	-0.7%
COLORADO	3.4%	6.2%	-2.0%
VEGAS	-1.5%	4.7%	-2.6%
EDMONTON	-1.6%	2.3%	-2.9%
CAROLINA	-2.9%	24.8%	-12.9%

Table 4. % Change in trap games vs similar non-trap games by fatigue level. At least one of the teams has to be playing back-to-back

FATIGUE LEVEL	GF	GA	GF%
FATIGUED TEAM	11.2%	-5.3%	10.5%
EQUAL FATIGUE	-0.4%	-2.2%	2.0%
FATIGUED OPPONENT	0.5%	10.5%	-4.8%

We also categorized situations based on fatigue, i.e., the number of days since a team last played a game. We focused on back-to-back situations as teams typically perform worse on the second half of a back-to-back. If neither team was in a back-to-back, we disregarded the game from our analysis since the difference in their fatigue levels was more marginal. For example, there was little difference between a team playing on three days of rest and another playing on four days of rest.

Table 4 showed a team playing back-to-back performed better in a trap game than they would otherwise if playing back-to-back. A stronger team playing back-to-back was less likely to underestimate their opponent given they recognize they are coming into an inherently disadvantageous situations. On the other hand, if a strong team faced a weak opponent who was doubly disadvantaged by playing back-to-back, the strong team's players may be more likely to underestimate their opponent.

In this section, we identified several trends in trap games for individual teams and fatigue levels that will be examined more rigorously in the following sections.

4. Permutation Tests

We created permutation testing frameworks to evaluate the previously identified results. We first built a one-sided test where the test statistic was the difference in the percentage of goals for (GF%) between trap games and similar non-trap

Table 5. One-Sided Permutation Tests in Trap Games over time. (1) Post-Lockout (2013–2019). (2) Post-Pandemic (2022–2025). (3) Pre-Lockout (2007–2012)

TEAM	P-VALUE	OBSERVED GF%	# TRAP GAMES
LOS ANGELES ¹	1.8%	-29.6%	14
CHICAGO ¹	2.3%	-25.5%	16
TORONTO ¹	5.6%	-18.4%	17
CAROLINA ²	8.7%	-12.9%	24
MINNESOTA ¹	9.7%	-16.0%	12
PITTSBURGH ³	18.4%	-11.0%	14
NASHVILLE ²	19.6%	-11.2%	11
NEW YORK ²	30.6%	-5.2%	13
ANAHEIM ¹	33.9%	-4.4%	22

games. The null hypothesis was no difference between trap games and similar non-trap games, with any difference due to random chance alone. The alternative hypothesis was that teams performed worse in trap games.

We then randomly permuted the percentages of goals for (GF%) in games across 10,000 simulations to derive simulated test statistics. The p-value was the number of simulated test statistics equal to the observed test statistic or further in the direction of the alternative.

Table 5 shows teams with the lowest p-values across various time periods. Applying a simple threshold of 5%, we rejected the null for two teams, the Los Angeles Kings and the Chicago Blackhawks during the Post-Lockout period from 2013–2019.

However, we tested multiply hypotheses in this framework so we needed to apply corrections for multiple hypotheses and control the Type I error rate. We applied both a Bonferroni correction as well as a less conservative Holm-Bonferroni correction given the 42 tests we conducted where the number of trap games exceeded 10. Across both correction methods, there were no statistically significant teams and thus we failed to reject the null across all tests.

We expanded this to a two-sided permutation testing framework when applying it to fatigue levels. We chose to make this test two-sided as the alternative hypothesis was that a team's fatigue level may cause the team to do better or worse in trap games, rather than focusing solely on whether it may do worse. This was due to the analysis we found in Table 4 of our previous section.

In the two-sided permutation test results shown in table 6, we see statistically significant outperformance by stronger teams who go into a trap game playing back-to-back compared to a similar back-to-back game. This was significant under not only a 5% significance level, but also under the stricter Bonferroni correction for multiple hypothesis test-

Table 6. Two-Sided Permutation Tests in Trap Games by Fatigue Level.

TEAM	P-VALUE	OBSERVED GF%	# TRAP GAMES
FATIGUED TEAM*	1.1%	10.5%	127
FATIGUED OPPONENT	11.5%	-4.8%	118
EQUAL FATIGUE	38.6%	1.97%	45

ing.

Therefore, our results showed that when a strong team was playing back-to-back against a weak opponent, it was better for that team to be in a trap game. Other fatigue levels were not significant in our two-sided permutation test.

Although we did not find any significant results for individual teams under a multiple testing framework, these corrections may be too conservative for this framework given a focus of this work was exploratory insight. Therefore, while we found it interesting that certain teams underperform in trap games, this was not sufficient to demonstrate broader systematic evidence of trap games. Indeed, our permutation testing framework demonstrated that a trap game was advantageous to stronger teams in back-to-back situations. These rigorous univariate results were then evaluated in multivariate settings.

5. Multivariate Models

The multivariate relationship between trap games and other factors such as fatigue and team strength is complex as seen in the permutation testing results. We defined our universe of games as those where a stronger team was facing a weaker team, as previously discussed. Let the dependent variable be the stronger team winning, i.e., collecting two points. This is a binary classification problem with relatively balanced distributions, i.e., the stronger team won around 57.4% of the time, thus alleviating the need for resampling.

We then considered independent variables to be whether it was a trap game, whether the stronger team was at home, the relative fatigue levels between both teams, and the performance over the season to date by both teams. These variables were the same as those previously defined. We applied one-hot encoding for categorical variables, z-score standardization, and winsorization of outliers beyond a threshold of three standard deviations. All independent variables with collinearity beyond 50% on a Pearson correlation basis were dropped in the final set of factors.

The train-validation split was based on season to avoid data leakage across time so all games in our validation set occurred after all games in our training set. We picked the

Table 7. Multivariate model performance measured by AUC.

MODEL	AUC
RANDOM FOREST	0.597
LOGISTIC REGRESSION	0.595
RIDGE REGULARIZED	0.595
LASSO REGULARIZED	0.594
GRADIENT BOOSTED	0.591
ELASTICNET REGULARIZED	0.591
BASELINE	0.477

Table 8. Important features in random forest model.

FEATURE	IMPORTANCE
HOME TEAM	0.278
OPPONENT AVG POINTS	0.261
TEAM AVG POINTS	0.182
FATIGUED TEAM	0.161
FATIGUED OPPONENT	0.091
TRAP GAME	0.020
POST-LOCKOUT SEASON	0.008

2023-2024 season as the cutoff point between the training/validation datasets in order to derive an approximate 86-14 split. The validation dataset consisted of all games during the 2023-2024 and 2024-2025 seasons.

The first model we created was a baseline derived from the distribution of the dependent variable. In addition, we trained logistic regressions, regularized models (Lasso, Ridge, ElasticNet), and ensemble machine learning models such as bagging and boosting. We used hyperparameter tuning to identify optimal parameters on the validation dataset with the objective of maximizing the AUC. See GitHub for a more detailed methodology outline and implementation.

Table 7 shows the performance of the models we built on an AUC basis, all of which exhibited modest improvement over the baseline.

More importantly for this work, we also observed the important factors across various models to determine if the trap game factor was important in the model's final decision-making.

In Table 8, we saw that the most important factors were whether the stronger team was at home and the strength of the weaker team. The strength of the stronger team and whether that stronger team was fatigued were of secondary importance. Whether it was a trap game had the second-lowest importance, demonstrating that a trap game was not important in the model's prediction.

These feature importances were consistent with what we observed in Table 9 for the logistic regression, which likewise

Table 9. Important features in logistic regression model.

FEATURE	COEFFICIENT	P-VALUE
HOME TEAM	0.321	<0.1%
OPPONENT AVG POINTS	-0.158	<0.1%
FATIGUED TEAM	-0.300	0.1%
TEAM AVG POINTS	0.076	1.3%
FATIGUED OPPONENT	0.224	1.9%
INTERCEPT	0.128	2.0%
TRAP GAME	0.048	53.9%
POST-LOCKOUT SEASON	-0.007	90.2%

did not observe a significant impact from whether it was a trap game. Similar conclusions on the non-importance of trap games were reached for all models and are available on GitHub in full detail.

Therefore, through multivariate models, we concluded that how stronger teams perform against weaker teams had more to do with factors such as their relative strengths and home-ice advantage rather than whether it was a trap game. When controlling for other factors in multivariate settings, trap games themselves were not tied to game outcomes.

6. Limitations

Our analysis did not control for travel and other factors which may impact how teams perform. However, travel was partially accounted for by fatigue and these missing factors should have limited impact on our conclusion that there was a lack of systematic evidence of trap games.

7. Conclusion

Our analysis showed that the trap game narrative is largely overstated. Across two decades of NHL games, strong teams facing weak opponents after a more challenging win performed as well or better than expected. While a few teams, such as the Los Angeles Kings and Chicago Blackhawks during the post-lockout era, showed evidence of underperformance, these conclusions did not hold after correcting for testing multiple hypotheses.

Trap games contribute minimally to predictive models when controlling for other factors such as home-ice advantage, relative team strengths, and fatigue. Therefore, the persistent narrative around trap games is a cognitive bias caused by over-fitting on memorable anecdotal upsets rather than systematic evidence.

References

NaturalStatTrick (2025). Natural stat trick.